Summary of the Ram Calculus Local-Smoothing Report

Main Results, Contributions Relative to the Source Paper, Quaternionic Methods, and Remaining Research Questions

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Abstract

This report summarizes the completed investigation of Ram Calculus and the October 25, 2026 OpenAI manuscript on Euclidean wave local smoothing. It distinguishes results already in that manuscript, exact reformulations using established mathematics, proved conditional additions in the Ram report, and proposals that have not been implemented or proved. The main additions are explicit Gaussian reconstruction budgets, quaternionic ordering and observation-loss estimates, boundary-conforming cofactor linearizations, residual-to-solution and residual-to-energy certificates, and an exact off-solution energy identity. The revised quaternionic TR60 provides ordered transform and real-block conventions; TR3 supplies a constraint-aware channel framework but does not itself enforce the nonlinear positive-Hessian condition. Manufactured examples and floating-point checks support the implementations of the stated identities. These contributions improve the transparency and potential verifiability of an approximation-based approach; they do not establish a sharper unrestricted local-smoothing estimate, a complete independent verification of the source proof, or a unique numerical advantage over equivalent conventional methods.

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1 Purpose, scope, and principal conclusions

The report being summarized is *Ram Calculus and Local Smoothing for the Euclidean Wave Equation: Quaternionic Structure, Auxiliary Dirichlet Problems, and Residual-Certified Reformulations* [1], referred to below as **the Ram report**. Its original source is the 59-page OpenAI manuscript *Local smoothing for the Euclidean wave equation*, dated October 25, 2026 [2], referred to as **the source paper**. The Ram report uses the supplied PDF and matching source at repository commit `42907c9e999cc00d6abe7a50bfc1023250edd1b9`. This summary explains the completed report; it is not a new literature survey or a claim to have performed a further full audit of the source.

The five principal conclusions are:

1. **Ram Calculus is substantively relevant.** Its window, moment, ordered quaternionic, frame, and residual methods can be connected precisely to several operations in the proof.
2. **The auxiliary PDE is not automatically inverted by a Ram partial differential operator.** It is a nonlinear quaternionic Moore-determinant Dirichlet equation. Its positive branch, boundary geometry, current conservation, and parameter dependence are essential, not removable notation.
3. **The strongest added result is an exact error-accounting identity for approximate potentials.** The energy balance acquires a determinant-residual times spacetime-Schur-complement term. This identifies what an approximate solver must control before it can support the proof's propagation step.
4. **Several additional bounds are proved under explicit hypotheses.** They quantify reconstruction stability, quaternionic ordering errors, elliptic solution errors, energy gaps, rank-one asymptotics, and weak planar inversion. Their validity does not depend on a claim that these classical ingredients originated with Ram Calculus.
5. **The unrestricted sharp-bound problem remains unchanged.** The investigation does not reduce the source's upper logarithmic loss, prove a faster general algorithm, or certify every analytic estimate needed for the source theorem.

What "new contribution" means here. An addition can be new *relative to the source paper's presentation* without being a new theorem in the worldwide literature. The Ram report adds a particular synthesis, application-specific identities, explicit certificates, and reproducible tests. It does not claim historical priority for spectral calculus, Taylor estimates, Schur complements, quaternionic Monge–Ampère theory, Lax–Milgram theory, or Galerkin approximation. The substantive contribution is making their connection to this proof and to the Ram corpus explicit and quantitatively usable.

2 What the original source paper supplies

2.1 The local-smoothing question

Let $n \geq 2$, $d = n - 1$, $D = -\mathrm{i}\nabla$, and $U(T) = e^{-\mathrm{i}T|D|}$. The half-wave solves $(\partial_T + \mathrm{i}|D|)u = 0$ and consequently the wave equation. Initially the data may be taken to be Schwartz functions on $\mathbb{R}^n$. The critical exponent is

$$p_{\mathrm{c}} = \frac{2n}{n-1}.$$

For a fixed smooth annular frequency cutoff φ and $N \geq 2$, the source asserts the critical estimate

$$\|U(T)\varphi(D/N)f\|_{L^{p_c}(\mathbb{R}^n\times[1,2])} \leq C[\log(2+N)]^{2/p_c} \|f\|_{L^{p_c}(\mathbb{R}^n)}, \tag{1}$$

with C independent of N and f. Its Sobolev consequence has the order

$$s_+(p)+\epsilon, \qquad s_+(p) = \max\{0, (n-1)/2 - n/p\}, \quad \epsilon > 0, \quad 2 \leq p < \infty.$$

The gain is in an L^p spacetime norm, not heat-like pointwise-in-time regularization of every rough input. A finite-dimensional reconstruction bound at one frequency cannot prove the all-frequency, all-input quantifier in (1).

The source's focusing example requires a critical logarithmic power at least $1/p_c$. Hence its upper power $2/p_c$ and this lower obstruction leave a gap. The Ram report does not close it. Both the claimed upper estimate and its far-reaching consequences remain source-dependent in the Ram report, rather than independently certified there.

2.2 The auxiliary PDE and proof mechanism

The source constructs a coherent quaternionic wave current $A^0 = \chi\chi^*$, regularizes it to a positive semidefinite conserved current A, and uses its positive time density $\mu = A_{TT}$. On a spatial fiber Ω_z, where z denotes the time and auxiliary parameters, it solves

$$D_u = H_{\text{fib}}(B_0 + u) > 0, \qquad \det_{\text{M}} D_u = \lambda\mu, \qquad u|_{\partial\Omega_z} = 0. \tag{2}$$

Here u is a real potential, the real spatial coordinates are periodic, $B_0 = \lambda|\Phi|^2$ is a rank-one Hessian background, and $\lambda = h^{-30n}$ for the wavelength parameter h. The Moore determinant is the real determinant polynomial appropriate to quaternionic Hermitian matrices. It is not the ordinary determinant formula applied to arbitrary noncommuting entries.

In the source convention, each quaternionic index has three real slots:

$$G_a(v) = \sum_{\alpha=0}^{2} e_\alpha \partial_{a\alpha} v, \qquad H_{ab}(v) = \sum_{\alpha,\beta=0}^{2} e_\alpha \overline{e_\beta}\, \partial_{a\alpha}\partial_{b\beta} v, \qquad (e_0, e_1, e_2) = (1, \mathbf{i}, \mathbf{j}).$$

For real v, $H(v)$ is Hermitian and $\operatorname{tr} H(v) = \Delta v$. The notation H_{fib} restricts this construction to the spatial fiber.

The source already supplies the following architecture:

1. Imaginary light-ray continuation produces a directional Gaussian and a coherent conserved current.
2. Quaternionic cofactor identities and the positive solution of (2) produce a scalar energy on the parameter space.
3. A special moving-boundary cancellation removes an indefinite full-spacetime determinant from that energy's Laplacian.
4. Upper and lower energy comparisons, together with two radial harmonic comparisons, produce the two-logarithm loss.
5. Transfer and interpolation give Euclidean wave estimates and further restriction, Kakeya, and dispersive consequences.

These constructions are *not inventions of the Ram report.* Likewise, the separate linear planar PDE in the source's Appendix D, its angularly restricted improvements, and its curved-pencil extension are credited to the source, not counted as new Ram results.

3 Contribution ledger: source results versus additions

Table 1 summarizes the attribution and evidence. Detailed section references refer to the Ram report, not to this summary.

Table 1: What is contributed, and what is not claimed.

Topic	Addition relative to the source presentation	Status and locator
Critical smoothing estimate	No stronger unrestricted exponent or logarithm is obtained.	Source claim; Ram report Sections 2, 17–18.
Gaussian continuation	Explicit spectral generator, finite moment remainder, inverse stability and noise budget.	Proved band-limited statements; Sections 5–6.
Quaternionic current	Ordered real blocks, mixed-slice commutator bound, physical metric and gauge-aware interpretation.	Exact identities and conditional error bounds; Section 10.
TR3 channels	Separation of projection loss from truncation loss, with a lower observation margin.	Application of an existing corpus theorem, not automatic nonlinear feasibility; Section 10.
Nonlinear auxiliary solve	Full cofactor Newton matrix and explicit coercivity bounds for conforming trial spaces.	Conditional result; Section 11.
Error in an approximate solution	Gradient, sup-norm, bracket, and energy-gap certificates.	Proved with positivity, boundary, and norm hypotheses; Section 12.
Approximate proof energy	Exact residual–Schur defect, plus explicit current-divergence and boundary obligations.	Main application-level addition; Section 13.
Rank-one background	Controlled reduced determinant, exact ball model, and explicit asymptotic error bounds.	Proved model results, not a general linearization; Section 14.
Planar solve	Weak-forcing extension and conforming polynomial test, built on the source's linear reduction.	Proved extension and executed manufactured example; Section 15.
Frames and propagation	Frequency-dependent reconstruction budget, compression-commutator margin, and window-dispersion test.	Proved conditional bounds and limitations; Section 16.
Accuracy and speed	No comparison showing superiority over an equivalent conventional discretization.	Not demonstrated; Sections 19–21.

4 Exact reformulations and their concrete benefits

4.1 The Ram partial differential operator: where it does apply

For a kernel operator written in displacement variables,

$$(Kf)(x) = \int h(x,\eta) f(x-\eta)\,\mathrm{d}\eta, \qquad S_\alpha(x) = \frac{(-1)^{|\alpha|}}{\alpha!}\int \eta^\alpha h(x,\eta)\,\mathrm{d}\eta,$$

integrated Taylor expansion gives

$$Kf = \sum_{|\alpha| \le M} S_\alpha \partial^\alpha f + R_M.$$

The Ram operator framework organizes this representation, its remainder, and all Leibniz terms when the coefficients vary. It can be applied to windows, integral kernels, and suitably specified linearized operators.

It does not make the nonlinear map $v \mapsto \det_{\mathrm{M}} H_{\mathrm{fib}}(B_0 + v)$ a fixed linear moment operator. An inverse also needs a domain, boundary conditions, closure of omitted derivatives or coefficients, and a stability bound. The finite inverse for a degree-lowering operator on polynomials is not an inverse theorem for the full elliptic PDE. The report further distinguishes window L^2 expansions from pointwise Taylor jets: a smooth, nonanalytic function is not determined by an unrestricted formal Taylor series. These distinctions prevent applying the right algebra to the wrong function space or equation.

4.2 Exact heat–Schrödinger representation at fixed null frequency

In null coordinates the source's plane waves take the form

$$u_{\kappa,v} = c_{\kappa,v} e^{\mathrm{i}\kappa(s_0 + v\cdot x - |v|^2 t/2)}, \qquad \kappa > 0.$$

At fixed κ, their transverse part solves $(2\mathrm{i}\kappa\partial_t + \Delta_x)u_\kappa = 0$ exactly. For real W and $y > 0$, the report identifies the coherent Gaussian window with a covariant heat semigroup:

$$F_{\kappa,y,W} = e^{\frac{y}{2\kappa}(\nabla_x - \mathrm{i}\kappa W)^2} u_\kappa, \tag{3}$$

$$\partial_y F_{\kappa,y,W} = \frac{1}{2\kappa}(\nabla_x - \mathrm{i}\kappa W)^2 F_{\kappa,y,W}, \qquad \partial_{W_j} F_{\kappa,y,W} = y(D_{x_j} - \kappa W_j) F_{\kappa,y,W}. \tag{4}$$

On a mode, the squared covariant derivative has symbol $-\kappa^2|v - W|^2$, which proves the identities. Parameter derivatives therefore give centered spectral moments without destroying phase coherence. This is an exact reformulation, not a paraxial approximation.

Benefit and limitation. Window size, moments, and phase evolution are expressed through compatible operators. But the source's additional normalization $(\kappa/y)^{d/2}$ contributes the zeroth-order term $-d/(2y)$ to the y equation. Ignoring that term is an error. Nor may different κ values be replaced by one fixed-κ PDE. The auxiliary determinant equation remains a separate problem.

4.3 A proved Gaussian approximation and inversion budget

For a unit vector ω, let

$$A_\omega = |D| - \omega \cdot D, \qquad M_\rho = e^{-\rho A_\omega}, \quad \rho \ge 0.$$

The symbol is nonnegative and equals $|\xi||\widehat{\xi} - \omega|^2/2$, so M_ρ is an L^2 contraction. On a band with $|\xi| \le C_0 N$ and $|\widehat{\xi} - \omega| \le \theta$, put $\delta = \rho C_0 N \theta^2/2$. The report proves

$$\left\| M_\rho f - \sum_{j=0}^{m} \frac{(-\rho A_\omega)^j}{j!} f \right\|_2 \le \frac{\delta^{m+1}}{(m+1)!} \|f\|_2, \tag{5}$$

$$\left\| M_\rho^{-1} \right\| \le e^{\delta} \quad \text{on this band.} \tag{6}$$

If an approximate multiplier has operator error $\epsilon < e^{-\delta}$ and the data error is at most η, its reconstruction error is at most

$$\frac{\eta + \epsilon \|f\|_2}{e^{-\delta} - \epsilon}. \tag{7}$$

For $\delta = 1, m = 6$, the truncation guarantee is $1/7!$, approximately 1.98413×10^{-4}, and the exact inverse amplification is at most e. This makes a formal window inversion quantitatively checkable. It does not regularize backward heat inversion on unrestricted frequencies, and an L^2 multiplier estimate alone does not prove the critical L^{p_c} estimate.

5 The precise role of quaternionic Ram Calculus and TR3

5.1 Why the quaternionic extension is genuinely relevant

The source already uses quaternionic matrices and a Moore determinant. The contribution of revised TR60 [3] is therefore not to introduce quaternions into the source proof. It provides an ordered transform and real-block language with which to represent those objects without silently changing their algebra.

In its corrected conventions, a historically named left-linear transform has its kernel *to the right* of the signal. A general two-sided transform is only real-linear. Quaternionic kernel coefficients, branch choices, derivative domains, and multiplication order must be fixed. The unconditional fixed-axis $q = 1$ Fourier Plancherel specialization cannot be promoted to a general inverse for arbitrary quaternion-order atoms without actual frame bounds. Similarly, a general quaternionic transform is not two independent complex transforms: the complex components usually couple.

Real blocks encode the exact ordered operations. With $L(p)$ and $R(q)$ representing left and right multiplication,

$$L(pq) = L(p)L(q), \qquad R(pq) = R(q)R(p), \qquad L(p)R(q) = R(q)L(p).$$

The reversal in the right-multiplication composition is essential. For a Hermitian matrix C, the real tensor

$$C^{\mathbb{R}}_{a\alpha,b\beta} = \mathrm{Re}(\overline{e_\alpha} C_{ab} e_\beta)$$

represents its contraction with the quaternionic Hessian exactly. In particular, cofactor quadratic forms become ordinary symmetric real quadratic forms, with all quaternionic information retained.

The physical norm also matters. A packed Hermitian quaternionic matrix has

$$\|C\|_F^2 = \sum_a C_{aa}^2 + 2 \sum_{a<b} |C_{ab}|^2.$$

Thus off-diagonal real components have weight two. Ignoring this weight changes the norm used to measure reconstruction errors.

5.2 TR3: separating channel loss from coefficient loss

TR3 [4] treats ordered hypercomplex channels on specified linear constraint spaces. In whitened real coordinates, let P project orthogonally onto the admissible input space, B_s be channel operators, P_s output orthogonal projections, and $F_{s,N}$ truncated analysis maps satisfying $F_{s,N}^* F_{s,N} = I - \mathcal{L}_s$,

$0 \preceq \mathcal{L}_s \preceq I$. For $f \in \operatorname{ran} P$, the observation map is $\mathcal{A}_N f = (F_{s,N} P_s B_s f)_s$, and

$$\mathcal{A}_N^* \mathcal{A}_N = P(G - D_{\text{loss}} - E_{\text{loss}})P\big|_{\operatorname{ran} P}, \tag{8}$$

$$G = \sum_s B_s^* B_s, \qquad D_{\text{loss}} = \sum_s B_s^*(I - P_s)B_s, \tag{9}$$

$$E_{\text{loss}} = \sum_s B_s^* P_s \mathcal{L}_s P_s B_s. \tag{10}$$

If V is an isometric synthesis of the chosen physical trial space, the sufficient margin

$$\gamma^2 = \lambda_{\min}(V^* G V) - \|V^* D_{\text{loss}} V\| - \|V^* E_{\text{loss}} V\| > 0$$

gives a reconstruction pseudoinverse norm at most $1/\gamma$.

Concrete advantage. One can identify whether lost information comes from removing physical channel components or retaining too few analysis coefficients. Complementary channels can restore observability: on imaginary quaternions the two projected maps $v \mapsto \mathbf{i} \times v$ and $v \mapsto \mathbf{j} \times v$ are individually singular, but their combined Gram is $\operatorname{diag}(1, 1, 2)$.

Essential limit. The framework concerns linear constraints. Hermitian symmetry can be included, but $D_v > 0$ and $\det_{\mathrm{M}} D_v = \lambda\mu$ are nonlinear conditions. A reconstructed matrix field also need not be the Hessian of one scalar potential. Spectral clipping to force positivity can destroy Hessian compatibility and cofactor conservation. TR3 does not eliminate these requirements.

5.3 The application-specific order-error result

In the wave current, factors from $\mathbb{C}_{\mathbf{k}}$ multiply embedded complex amplitudes from a generally different slice. For a unit $e \in \mathbb{C}_{\mathbf{k}}$ and real a, b, α, β, the report derives

$$[a + \mathbf{k}b, \alpha + e\mathbf{i}\beta] = 2b\beta e\mathbf{j}, \qquad |[a + \mathbf{k}b, \alpha + e\mathbf{i}\beta]| = 2|b\beta|. \tag{11}$$

Consequently, reversing these factors in a coherent sum can change its amplitude. For a current factor χ and an approximation $\widetilde{\chi}$, the report proves

$$\|\chi\chi^* - \widetilde{\chi}\widetilde{\chi}^*\|_F \le (\|\chi\| + \|\widetilde{\chi}\|)\,\|\chi - \widetilde{\chi}\|\,. \tag{12}$$

The amplitude error caused by modewise reversal is bounded by the sum of the commutator magnitudes in (11), weighted by the real Gaussian factors and the imaginary parts of the polarization columns. This turns multiplication order from a verbal warning into an explicit error budget.

Not every amplitude change is observable in the current. A common right unit quaternion r gives $(\chi r)(\chi r)^* = \chi\chi^*$ exactly. This is a gauge ambiguity. TR3's linear observation results must be used on an appropriate linearization or lift with that gauge accounted for, rather than being applied without qualification to the nonlinear map $\chi \mapsto \chi\chi^*$. Nor may a coherent density be replaced by a sum of independent mode energies: $|1 + 1|^2 = 4$ and $|1 - 1|^2 = 0$, whereas both corresponding modewise energy sums equal two.

6 Proved improvements in solving and certifying the auxiliary PDE

These are improvements in the *explicit quantitative control of a candidate approximation.* They are not stronger general existence theorems for the source's PDE and do not establish faster convergence.

6.1 Boundary-aware Newton equations and coercivity

For a real admissible potential v, define

$$D_v = H_{\mathrm{fib}}(B_0 + v), \quad \mathcal{R}(v) = \det_{\mathrm{M}} D_v - \lambda\mu, \quad \mathcal{L}(v) = \log \det_{\mathrm{M}} D_v - \log(\lambda\mu).$$

The exact linearizations are

$$D\mathcal{R}(v)[w] = \mathcal{C}(D_v) : H_{\mathrm{fib}} w, \qquad D\mathcal{L}(v)[w] = D_v^{-1} : H_{\mathrm{fib}} w, \qquad A : B = \operatorname{Re}\operatorname{tr}(AB).$$

No additional zeroth-order $+w$ term is present. For real trial functions ϕ_j satisfying the periodic and zero Dirichlet conditions, the determinant Newton matrix is

$$K_{ij} = \int_{\Omega_z} \operatorname{Re}\{G_{\mathrm{fib}}(\phi_i)^* \mathcal{C}(D_v) G_{\mathrm{fib}}(\phi_j)\}\, \mathrm{d}\zeta.$$

If $cI \preceq D_v \preceq CI$, the proved comparison is

$$c^{n-1} K_\Delta \preceq K \preceq C^{n-1} K_\Delta, \qquad (K_\Delta)_{ij} = \int \nabla\phi_i \cdot \nabla\phi_j. \tag{13}$$

The cofactor eigenvalues are products of $n-1$ Hessian eigenvalues, which explains the powers. A step $v + tw$ stays positive if $t \left\| H_{\mathrm{fib}} w \right\|_{\infty,\mathrm{op}} < c$. These statements justify a conforming Ram-panel implementation, but they hold for conventional conforming bases as well. Quadrature, conditioning, and nonlinear convergence still need separate control.

6.2 Residual-to-solution and barrier bounds

Let $V = H^1_{D,\mathrm{per}}(\Omega_z)$ have norm $\|w\|_V = \|\nabla w\|_2$, and let V^* have the dual norm. For an exact solution u and a trial v with the same boundary data, assume *both* $D_u, D_v \succeq cI$. Then

$$\|v - u\|_V \le c^{-(n-1)} \left\| \mathcal{R}(v) \right\|_{V^*}. \tag{14}$$

If $H_{\mathrm{fib}} r \succeq \kappa I$ for a defining function $r < 0$ with $r = 0$ on the boundary, the pointwise certificate is

$$|v - u| \le \frac{\|\mathcal{R}(v)\|_\infty}{n c^{n-1} \kappa}(-r). \tag{15}$$

The proof linearizes the determinant along the positive segment between u and v. Its mean cofactor is at least $c^{n-1} I$ and is divergence-free in real form. Testing with $v - u$ proves (14); comparison with multiples of r proves (15).

There is also a log-residual sup-norm estimate when $0 < D_u, D_v \preceq CI$:

$$|v - u| \le \frac{C \left\| \mathcal{L}(v) \right\|_\infty}{n\kappa}(-r).$$

Its mean-inverse operator is used in nondivergence form; an inverse Hessian is not generally divergence-free.

Why the conditions matter. Positivity of a trial Hessian alone does not provide the required lower bound on the unknown exact Hessian. The report gives an explicit ball example where using only the trial margin predicts an error at most 0.1, but the actual error is 0.125. A separate negative control, $\mathrm{diag}(-1, -1, 1)$, has positive determinant but is not positive definite. Neither a small sampled determinant residual nor a positive determinant selects the elliptic branch.

An alternative is to certify explicit sub- and supersolutions. If $v \pm \delta r$ are admissible and their determinants bracket $\lambda\mu$ in the correct order throughout the fiber, comparison gives $v + \delta r \le u \le v - \delta r$. This replaces an unknown endpoint bound by a verifiable continuum bracket, not by checking finitely many sample points alone.

6.3 A residual-to-energy certificate

Define the source's variational energy on a fixed fiber by

$$E_z(v) = \lambda^{-1} \int_{\Omega_z} v \int_0^1 \det_{\mathrm{M}} H_{\mathrm{fib}}(B_0 + \theta v)\, \mathrm{d}\theta\, \mathrm{d}\zeta, \tag{16}$$

$$\mathcal{J}_z(v) = E_z(v) - \int_{\Omega_z} v\mu\, \mathrm{d}\zeta. \tag{17}$$

Its derivative is the normalized residual $R_v = \lambda^{-1} \det_{\mathrm{M}} D_v - \mu$. If $\lambda^{-1}\mathcal{C}(D_{u+t(v-u)}) \succeq \kappa_0 I$ along the segment, and $r_* = \|R_v\|_{V^*}$, the report proves

$$\|v-u\|_V \le r_*/\kappa_0, \qquad \frac{\kappa_0}{2}\|v-u\|_V^2 \le \mathcal{J}_z(u) - \mathcal{J}_z(v) \le \frac{r_*^2}{2\kappa_0}. \tag{18}$$

This follows from strong concavity of $\mathcal{J}_z$ on the segment. It quantitatively relates the residual to an error in the variational energy, not just an error in the potential. The constants may worsen as $h \to 0$; uniformity is not supplied by the certificate itself.

6.4 The role of the nonlinear extension TR4

TR4 [7] explains why a nonlinear calculation must evaluate the decoded physical residual, including omitted coefficient directions and tails. If an encoder depends on v, differentiating its encoded residual includes the derivative of the encoder as well as the PDE Jacobian. Moving-fiber charts also contribute Jacobian and boundary derivatives. A certified contraction requires a true inverse on the full residual space, not a finite-rank operator that annihilates an unobserved residual. These are requirements for a valid solver and validation proof, not an implemented full solve in the present report.

7 The central added identity: energy away from an exact solution

This is the most direct addition to the source's exact-solution argument. It answers: *if the auxiliary PDE is solved only approximately, what error enters the proof energy?*

Fix the source's current, density, moving domain, rank-one background, and parameter-independent λ. Let v be a smooth family with zero trace on the full moving wall and positive spatial block D_v. Write its full spacetime Hessian as

$$H(B_0+v) = \begin{pmatrix} \ell & c^* \\ c & D_v \end{pmatrix} = \sigma_v e_T e_T^* + \Sigma_v, \quad \sigma_v = \ell - c^* D_v^{-1} c, \quad \Sigma_v \succeq 0.$$

The scalar σ_v is the Schur complement and need not be positive. Set $\mathcal{F}_v = \mathcal{J}_z(v)$ and let $\rho_v = (v/r)|_{r=0}$. The report derives the exact identity

$$\boxed{\Delta_z \mathcal{F}_v + \int_{\Omega_z} A : \Sigma_v\, \mathrm{d}\zeta = \mathcal{B}_v(z) + \int_{\Omega_z} R_v \sigma_v\, \mathrm{d}\zeta,} \tag{19}$$

where $\mathcal{B}_v$ is the source's moving-wall flux with v:

$$\mathcal{B}_v(z) = \int \rho_v\, G(r)^* A_{\mathrm{w}} G(r)\, \delta(r)\, \mathrm{d}\zeta.$$

Here A_{w} is the wave-current portion of the regularized current, $G(r)$ is the *full* gradient, and $\delta(r)\,\mathrm{d}\zeta = \mathrm{d}S_{\mathrm{fib}}/|\nabla_{\mathrm{fib}} r|$.

7.1 Why the identity holds

The current Green identity gives

$$\Delta_z \int (-v)\mu = \mathcal{B}_v - \int A : \Sigma_v - \int \mu\sigma_v.$$

The source's cofactor calculation continues to give $\Delta_z E_z(v) = \lambda^{-1} \int \det_{\mathrm{M}} H(B_0 + v)$ even away from a solution: the cofactor wall cancellation uses zero trace, the shared null-line geometry, and the rank-one background, not the determinant equation. Finally,

$$\lambda^{-1} \det_{\mathrm{M}} H(B_0 + v) = (\mu + R_v)\sigma_v.$$

Adding the two formulas leaves precisely the last term in (19). At an exact solution, $R_v = 0$, and the source's energy identity is recovered.

7.2 The new information and its limits

The bound

$$\left| \int R_v \sigma_v \right| \le \|R_v\|_2 \, \|\sigma_v\|_2$$

shows why a small elliptic determinant residual is insufficient by itself. The error is multiplied by a factor containing parameter derivatives and D_v^{-1}. A solver that computes an accurate potential separately on each fiber may still fail to control this spacetime factor. Moreover, positivity of D_v alone does not imply positivity of the trial energy $\mathcal{F}_v$. The exact-solution comparisons cannot simply be reused for an approximate family.

If the computed current has real column divergence $b = \operatorname{div} A^{\mathbb{R}} \neq 0$, Green's identity acquires

$$D_b(\eta) = \int b \cdot (\eta \nabla v - v \nabla \eta).$$

It enters with a plus sign in the identity for $\int v A_{TT} \Delta\eta$, but with a *minus* sign in the energy term based on $-v\mu$. Errors in tangency, background annihilation, or boundary trace generate further defects. Equation (19) is therefore an exact answer under named hypotheses, not a blanket guarantee for arbitrary discretizations.

Contribution relative to the source. The source uses exact auxiliary solutions to build its energy argument. The Ram report makes the surviving error for a trial solution explicit and links it to a future validated computation. That is a substantive extension of the error analysis of the proof mechanism, not a replacement proof of the critical estimate.

8 Other proved extensions and diagnostic bounds

8.1 Rank-one reduction and a fully solvable comparison model

For quaternionic Hermitian H, a unit column e, and real λ,

$$\det_{\mathrm{M}}(H + \lambda e e^*) = \det_{\mathrm{M}} H + \lambda \det_{\mathrm{M}}(H|_{e^\perp}). \tag{20}$$

This follows from the Moore Schur formula and polynomial continuation. Dividing by $\lambda > 0$ exposes a degree-$(n-1)$ reduced determinant and a correction $\lambda^{-1} \det_{\mathrm{M}} H$. If $\|H\|_{\mathrm{op}} \le M$, the correction

has absolute size at most M^n/λ. The hypothesis is essential: a Hessian growing with λ may defeat the approximation.

On an imaginary unit ball with $r = |P|^2 - 1$, background $B_0 = \lambda|P_1|^2/4$, and constant density $\mu > 0$, the exact zero-Dirichlet solution is $u = \alpha r/4$, where

$$\alpha^{n-1}(\lambda + \alpha) = \lambda\mu, \qquad \alpha > 0.$$

Put $d = n - 1$ and $\alpha_0 = \mu^{1/d}$. The report gives

$$\alpha_0(1 + \alpha_0/\lambda)^{-1/d} \le \alpha \le \alpha_0, \tag{21}$$

$$0 \le \alpha_0 - \alpha \le \frac{\alpha^2}{d\lambda} \le \frac{\alpha_0^2}{d\lambda}. \tag{22}$$

This is a rigorously controlled example of the rank-one scaling. For $n = 3$, $\mu = 2$, $\lambda = 100$, the recorded root is approximately 1.4043864384, and the error from $\sqrt{2}$ is $0.00982712394 < 0.01$. For $n \ge 3$ the reduced determinant remains nonlinear. Neither (20) nor the model solves the source's moving-fiber family in general.

8.2 Planar weak forcing and conforming approximation

The source already reduces the planar alternative to disk equations of the form $A_k w_k = f_k$, with $A_k = -\Delta + V_k$, $V_k \ge 0$, and zero Dirichlet data. Building on this reduction, the Ram report extends the fixed-mode problem to $f_k \in H^{-1}$ and the modewise L^2 inverse to square-summable divided forcing. For bounded V_k at each mode, with the gradient norm on H_0^1 and its dual,

$$\left\| A_k^{-1} \right\|_{L^2 \to L^2} \le \lambda_1^{-1}, \tag{23}$$

$$\|\nabla(w_k - \widetilde{w}_k)\|_2 \le \|f_k - A_k \widetilde{w}_k\|_{H^{-1}}. \tag{24}$$

Here λ_1 is the first disk Dirichlet eigenvalue. These are Lax–Milgram and Poincaré consequences with useful uniformity in the nonnegative mode potential. Extending the forcing for this linear auxiliary equation does not weaken the assumptions of the whole wave theorem: higher parameter derivatives and flux control are additional.

8.3 TR1, TR2, and the cost of reconstructing the wave

TR1 [5] supplies an actual L^p patch Banach frame with finite overlap and a square-function coefficient norm, not an arbitrary identification of coefficients with unweighted ℓ^p. The remaining task is to bound the encoded wave evolution uniformly in frequency. Reconstruction stability by itself is not local smoothing.

For any annular reconstruction error e_N, the report proves without assuming the contested critical estimate that

$$\|U(T)e_N\|_{L^{p_c}_{T,x}} \le CN^{1/2} \|e_N\|_2 \tag{25}$$

on a unit time interval, by Bernstein and L^2 unitarity. Thus a fixed relative L^2 error is not automatically harmless at high frequency. Equation (25) supplies an explicit accuracy budget for a frame or compressed reconstruction.

TR2 [6] also motivates a wave-specific compression bound. On a bounded band let $H = |D|$, let Q be a contraction with leakage $\mathcal{E} = I - Q^*Q$, and let V be an isometric finite-trial synthesis. For

$K_T = V^* e^{\mathrm{i}TH} \mathcal{E} e^{-\mathrm{i}TH} V$,

$$(Qe^{-\mathrm{i}TH}V)^*(Qe^{-\mathrm{i}TH}V) = I - K_T, \qquad \|K_T - K_{T_0}\| \le |T - T_0|\,\|[H, \mathcal{E}]\|\,.$$

This describes loss caused by compression after propagation, not by the uncompressed unitary wave. It is an adaptation of TR2's commutator method, not an identification of $|D|$ with its logarithmic generators.

Finally, replacing the wave symbol by a window-modified dispersion relation changes the PDE. In the report's window example, $\omega_b(\xi) = c|\xi|\{1 - (b\xi)^2/40 + O((b\xi)^4)\}$. Small relative symbol error $bN \to 0$ does not guarantee small absolute phase error at fixed nonzero time; a sufficient condition in that regime is $b^2N^3 \to 0$. This is a useful consistency test for proposed Ram-window wave approximations, not an improved estimate for a different propagator disguised as the original one.

9 What the executed tests establish

The recorded experiments use NumPy 2.3.5, seed 20261026, in `local-smoothing-support/check_extensions.py`; numerical outputs are in `local-smoothing-support/extension-check-results.json`. Table 2 summarizes representative values from that run. They are not new experiments performed solely for this summary.

Table 2: Algebraic and manufactured-solution checks already executed.

Check	Cases	Recorded result
Quaternionic rank-one determinant	120	Maximum relative discrepancy 1.78×10^{-15}.
Moore determinant directional derivative	120	Maximum scaled discrepancy 3.62×10^{-10}.
Quaternionic Schur identity	120	Maximum relative discrepancy 9.97×10^{-16}.
Directional Gaussian identity	500	Maximum absolute discrepancy 3.33×10^{-16}.
Fixed-κ Schrödinger relation	24 modes	Residual 6.99×10^{-16}.
Mixed-slice commutator	100	Maximum block discrepancy 7.02×10^{-16}.
Common right-phase current invariance	100	Maximum block discrepancy 8.39×10^{-15}.
Current-error inequality	100	Largest observed left/right ratio $0.863189 < 1$.
Positive-root ball model	9	All tested residuals and bounds passed.

The complex representation used in these checks is faithful to quaternion multiplication. The positive square root of its complex determinant was used only for *positive* Hermitian matrices; it does not recover the sign of an indefinite Moore determinant. The derivative test used Richardson-corrected finite differences, which explains its larger floating-point discrepancy. None of these finite checks substitutes for an all-input proof or interval certification.

9.1 What the polynomial example demonstrates

The manufactured unit-disk equation is

$$A = -\Delta + 3, \qquad u_*(r) = (1 - r^2)(1 + 0.4r^2 + 0.2r^4), \qquad f = Au_*.$$

With conforming basis $\phi_j = (1 - r^2)r^{2j}$, the results are:

Highest j	L^2 error	L^2 residual	$\operatorname{cond}_2 K$
0	1.10741×10^{-1}	6.49165	1.000
1	1.01778×10^{-2}	1.30713	7.588
2	1.92×10^{-16}	6.41×10^{-15}	92.981

Enlarging the space improves accuracy on this model, while its matrix condition number increases. The final near-zero error occurs because u_* lies exactly in the degree-two trial space. The conservative certificate $\|u_* - \widetilde{u}\|_2 \le \|f - A\widetilde{u}\|_2 / 3$ holds since $A \succeq 3I$.

These basis functions may be organized as Ram panels, but an ordinary Galerkin method using the same functions, integrals, and boundary conditions has the same approximation power. The test does not compare two different methods or demonstrate a uniquely Ram-derived accuracy gain. In particular, the actual variable-coefficient planar family, the higher-dimensional nonlinear moving-fiber problem, and a complete local-smoothing computation were not solved in these experiments.

10 What remains open, and which improvements are only proposed

10.1 The sharp unrestricted logarithm

The source's scalar radial comparison incurs one logarithm on an outward passage and one on the return. Its counterexample is a nonnegative supersolution whose inner time trace remains nontrivial, while the returning value is only of order $[\log(1/b)]^{-2}$. It prevents a universal one-logarithm comparison for the same class of scalar supersolutions. An invertible basis or coordinate change cannot remove that counterexample.

A possible improvement would have to exploit additional identities satisfied by *actual wave-generated energies*, excluding this larger supersolution class. This is a proposed research direction. The source already obtains better estimates under angular restrictions; those are restricted-input results and are not a new unrestricted Ram improvement. No new general Kakeya, restriction, Bochner–Riesz, or Schrödinger theorem is proved in the Ram report.

10.2 A validated computation for the actual auxiliary family

The most concrete next implementation would use the actual W-dependent planar disk operators, retain their boundary conditions, and certify Fourier tails, parameter derivatives, and wall flux. For $n \ge 3$, it would additionally require continuum positivity, nonlinear existence or a validated solution bracket, and full resolved–tail inversion. Constants must be tracked as $h \to 0$. A converged numerical solution at one h is not the scale-uniform analytic statement required by the source.

The defect in (19) gives a useful success criterion: bound the residual–Schur product and all geometric/current defects in norms compatible with the propagation argument. This is more demanding than reporting a small residual at collocation points, but it is also a clearer target for future computational work.

10.3 Compression, wider equations, and efficiency

Other proposals include:

1. Gauge-aware quaternionic compression using TR3, with separate certificates for current error, divergence, tangency, and boundary flux.
2. Frequency-uniform L^p bounds for the wave operator in TR1's actual coefficient norm, rather than finite-cutoff matrix experiments.
3. TR4-based adaptive trial spaces with certified unresolved tails, compared against conventional spectral and finite-element methods on the same physical PDE and error tolerance.
4. Extensions to rough coefficients or more general characteristic sheets, with coefficient regularity, curvature, and spectral-gap assumptions stated and proved rather than discarded by representation.

These are not completed results. Runtime, memory, conditioning, and accuracy improvements require controlled benchmarks; new asymptotic claims require analysis. Formal or independent verification of all source steps is another separate undertaking. The bounded audit found no demonstrated fatal error in the particular algebraic and comparison steps it checked, but this is not a certificate for the complete proof.

11 Overall assessment: advantages, limitations, and attribution

Advantages established by the report. The report provides a more explicit account of how representation, noncommutative order, physical norms, boundary conditions, and residuals interact. It derives quantitative bounds that can be checked under named hypotheses and identifies concrete failure modes: missing window normalization terms, destroyed coherence, wrong quaternion order, unobservable gauge changes, false nonlinear branch selection, and uncontrolled parameter derivatives. The exact off-solution identity connects these computational concerns directly to the proof's energy mechanism. This is substantive analytical and methodological progress.

Costs and limitations. Ordered quaternionic blocks and constrained channels require additional bookkeeping. Positivity and Hessian compatibility are nonlinear, tails must be retained, and backward Gaussian inversion can be ill-conditioned. Strong local certificates may have poor wavelength-dependent constants. An equivalent Ram representation does not by itself improve the underlying mathematical approximation space or remove a sharpness obstruction.

What is specifically Ram-based. The relevant Ram corpus gives a coherent framework connecting window moments, ordered hypercomplex transforms, compatible trial spaces, physical metrics, observation losses, and nonlinear residual validation. Applying that framework to the source exposes useful links and supplies the stated application-specific calculations. The proved mathematical inequalities are nevertheless available to a conventional analyst who uses the same hypotheses and calculations. No test in the report establishes that the benefit is inaccessible to conventional methods.

Bottom line. The report makes the auxiliary-PDE approach more explicit, testable, and suitable for future validated approximation. It adds proved conditional error control and a particularly

informative off-solution energy formula. It does *not* claim a sharper unrestricted local-smoothing bound or an independently completed verification of the source's theorem. The promising next step is to turn these certificates into a full, scale-aware validated computation and then determine whether extra wave structure yields a genuine sharper estimate.

References

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